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A non-vacuum approach to classify deformation in hcp alloys combining polarized light microscopy, panoptic segmentation, and fundamental crystallography

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A B S T R A C T

Twinning, where adjacent microstructural grains share a common crystallographic plane, is an important accommodation of plastic deformation within many crystalline materials. Although twins are readily visible using optical microscopy, their crystallographic classification typically requires vacuum-based techniques such as electron backscatter diffraction (EBSD). Here, we demonstrate for the first time that an accessible, non-vacuum approach combining polarized light microscopy (PLM) with an open-source panoptic segmentation model (YOLO) can identify and classify deformation twins in Ti alloys. This work provides the first evidence that PLM integrated with machine learning can be used for twin identification and classification while operating entirely outside a vacuum environment. Because PLM + YOLO detects twins based on morphological and optical features, rather than the precise pixel-level misorientations required by EBSD, the approach captures twins that are poorly developed, blurred, or below EBSD's angular resolution. As a result, the YOLO models achieve up to twice the twin detection rate of EBSD at low strain levels ($\varepsilon \leq 1.0\%$). Regarding twin classification, the work demonstrates that PLM provides sufficient crystallographic information to distinguish tension from compression twins in a subset of cases. However, classification is subject to both theoretical limitations and engineering limitations. In the present implementation, approximately 5% of unclassified twins are attributable to theoretical constraints, while the majority of unclassified cases reflect current engineering limitations that are addressable through improved calibration, segmentation models, and parent-twin identification algorithms. This work therefore establishes PLM integrated with machine learning as a promising foundation for real-time monitoring of microstructural evolution during manufacturing, offering a low-cost solution for materials characterisation.

1. Introduction

Titanium alloys are widely used in lightweight high-performance applications. For example, Ti-alloys typically account for approximately 30% of the structural weight of a modern gas turbine aero engine [1]. Their high corrosion resistance, specific strength, and excellent fatigue resistance at temperatures up to 600 °C make them suitable for a large range of applications inside the engine like fan discs and blades, compressor discs, and static components [2]. During use, a range of stress, temperature and frequency combinations can alter the life span of titanium alloys. As part integrity is critical for aero-engine components, the ability to directly monitor and control twin activity, an excellent indicator of local deformation processes, could reduce parts premature failure or cracking and increase quality assurance. The hexagonal close-packed (hcp) structure of Ti is problematic as its C/A (axial) ratio (1.587), is less than the ideal axial ratio of 1.633 [3] (Fig. 5a.). As such, prismatic glide or pyramidal glides ($\langle A \rangle$ and $\langle C + A \rangle$) are favoured over basal glide [4]. Alongside these deformation modes, deformation twinning is also common in α-Ti alloys. The two most common twinning systems being $\{11\bar{2}2\}\langle 11\bar{2}3\rangle$ contraction twins (CTs) and the $\{10\bar{1}2\}\langle 10\bar{1}\bar{1}\rangle$ tension twins (ETs) [5] (Fig. 5a.). Twinning generally follows a three-step process: nucleation at stressed interfaces, activation once a small volume (few nm^3) is reached, and transverse propagation across the grain [6,7].

Several characterisation strategies have been explored to quantify twin activity within materials. Techniques commonly used to measure the twin activity include optical microscopy (OM), transmission electron microscopy (TEM), and electron backscatter diffraction (EBSD). Probable twins can be identified from optical micrographs using grain shape as the only criteria. At low strains, twins can be relatively easy to segment with based on their characteristic morphologies (i.e. geometric information), but it becomes more difficult to differentiate between parent grains and deformation twins at higher strains [8]. Hence, measuring twin density is often difficult and imprecise. On the other

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hand, TEM can give precise crystallographic information (i.e. physical information) about a small number of twins within a field of view [9]. Nevertheless, labour-intensive TEM sample preparation, and the requirement for electron transparent samples, inhibits acquisition of statistical information across samples larger than several hundred microns. For precise crystallographic information over larger areas, electron backscatter diffraction (EBSD) stands out as a best technique to obtain large dataset, especially using high-speed CMOS cameras that are now commonplace [10]. EBSD can extract information pertaining to absolute and relative crystal orientation and phase information on a pixel-wise basis [11]. To expand the capacity of EBSD, semi and fully automated methods have been developed to perform larger-scale quantification of twinning with good accuracy using crystallographic information. The four dominant twin types for hcp Ti are reported in Table 1. Different criteria are usually setup to identify different twins in a sample. Firstly, misorientations are appraised to determine whether a given boundary corresponds to a twin boundary of a given type [12]. This first criterion can lead to errors, and it is common that 'chance' boundary segments will meet a twin criterion while the grain map suggests otherwise. Hence, additional criteria can be applied to avoid these errors: measurement of the effective Schmid factor [13], a voting scheme to measure the parents of the twins [14], or extraction of line segments to determine the parent from the child [15].

While EBSD can be used to extract local parent-twin information within prepared samples, EBSD requires a degree of skill in both sample preparation and operation. Furthermore, EBSD must be performed in a high vacuum environment [16]. Polarized light microscopy (PLM) is a type of optical microscopy, and can be used to capture grain morphology information, but also grain orientation along the optical axis for optically anisotropic materials such as α-Ti, using reflected light intensity [17]. Hence, PLM can be applied identify two Euler angles along the optical axis (azimuth, φ, and inclination, θ). PLM does not need a vacuum to operate. Moreover, twin systems that share a twinning plane do not have a change in rotation along the optical axis. Therefore, even if PLM could not be as performant as vacuum methods like EBSD, PLM could be a possible candidate for twin 'classification', once twins and parent grains have been segmented. McCabe et al. were the first ones to introduce computational methods to segment both twins and grains in PLM [18]. They were able to identify thin twins that EBSD could missed due to spatial resolution. Nevertheless, like other characterisation methods, PLM has some practical limitations. Firstly, PLM is sensitive to noise and requires flat, polished surfaces [19]. Secondly, the reflected intensity requires a precise calibration to avoid errors for the inclination angle [18]. Hence, computational methods have been developed to improve grain segmentation [20] but also for better identification of the Euler angles [18]. Machine learning (ML) models have recently been introduced to facilitate grain segmentation [21].

In this study, PLM is used to identify twinning in a pure hcp titanium sample using two image panoptic segmentation models based on YOLOv8-seg. Firstly, a model to identify parent grains already described in previous work [21]. Then, a second model to identify the twins that will be presented in this work. The observations are compared with an EBSD dataset and discussed. The PLM images acquired from α-Ti samples were prepared using electrochemical jet machining (EJM) [20], but the general approach could be deployed to different etched surfaces independent of the etching routine or material, providing it is optically anisotropic. Finally, the study discusses the opportunities to identify and classify twins pushing forward the capabilities of PLM for material characterisation.

Table 1
Twin type parameters for α-titanium (see Fig. 2c).

Type	K_1 Plane	η_1 Direction	Axis-Angle	Colour
T1	$\{10\bar{1}2\}$	$\langle 10\bar{1}\bar{1}\rangle$	$\langle 11\bar{2}0\rangle$ – 84.78 deg	lime
T2	$\{11\bar{2}1\}$	$\langle 11\bar{2}\bar{6}\rangle$	$\langle 10\bar{1}0\rangle$ – 35.10 deg	lime
C1	$\{11\bar{2}2\}$	$\langle 11\bar{2}\bar{3}\rangle$	$\langle 10\bar{1}0\rangle$ – 64.62 deg	blue
C2	$\{10\bar{1}1\}$	$\langle 10\bar{1}2\rangle$	$\langle 11\bar{2}0\rangle$ – 57.42 deg	blue

2. Methodology

2.1. Experimental procedure

Rolled polycrystalline samples of commercially pure titanium (grade 1, Ti CP1) of composition: C 0.1, Fe 0.2, N 0.03, O 0.18, H 0.015 (wt%), Ti balance, were used in this study. In total, six samples were prepared to train the model and one sample to test the model. Surfaces were ground and polished (up to 1 μm diamond) prior to electrochemical jet machining (EJM). The electrolyte composition was a mixture of choline chloride and ethylene glycol in a 1:4 mass ratio into which NaBr (3 wt%) was dissolved. Measurement areas (10 mm long by 2 mm wide) were selectively etched on the samples using this EJM system. All measurement regions were prepared with the electrolyte temperature and an environment temperature maintained at 25 °C. The EJM system consists of a constant current power supply and 3-axis positioning system controlled by a motion controller, previously described [22]. The electrolyte jet speed is controlled with a gear pump. The jet inner diameter, electrolyte jet velocity and the inter-electrode gap were kept constant during the machining, respectively at 1.09 mm, 12.7 m/s and 0.5 mm. The fixed parameters have been described previously [20] and the parameters were fixed with a current of 150 mA and a jet translation speed of 1.08 mm/s.

To investigate the influence of defects and twinning in pure titanium, a Deben microtest 5 kN testing stage was used (Fig. 5b.). The stage can operate in a tensile or compression mode with a force of up to 5 kN. The stage was positioned below an Alicona G4 InfiniteFocus microscope to capture images of the samples using PLM. The images were captured after a cycle of sample loading/unloading using a 20× objective leading to an effective image resolution of 0.44 μm/pixel. After each unloading, the sample were mounted in a Zeiss Crossbeam 550 electron microscope, where electron backscatter diffraction (EBSD) datasets were collected using a CMOS detector. All EBSD datasets were acquired with an emission current of 15 mA, an accelerating voltage of 15 kV and a step size and a pixel size of 0.60 μm, coarser than the optical resolution. There were seven samples in total: six samples were used to train the model and one to test the model. The samples used for training had strain levels ranging from 0 to 1.0% of tension or compression true strain. The test sample was a tensile sample and was loaded and unloaded three times with the different values of strains after each unloading being 0%, 0.6%, 1.2% and 2.4%.

2.2. Dataset construction for the training YOLOv8-seg twin model

To train a model to recognize twinning regions, datasets were constructed from PLM images of the prepared surfaces according to Fig. 1. PLM images are acquired at different rotation angles, ω (0° - 180°, $\Delta\omega = 10°$) and converted to grayscale. For a given region, four unique pseudo-colour images were composed retaining a constant rotation angle ($\Delta\omega = 40°$) between each colour channel but switching the order and the initial image. After pseudo-colour image generation, twinning regions were manually labelled to generate the dataset and superposed to the acquired images to build the ground truth (Fig. 1a.). Then, the different images and associated labels were cropped and split into the validation and training datasets. In total, 3144 images were assigned for training, and 792 images were assigned for validation. The YOLOv8-seg twin model was trained from the YOLOv8-seg grain model until no further improvements were reported for precision, the accuracy of positive predictions, and recall, the model's ability to find all positive instances. The trained YOLOv8-seg twin model was verified using test data consisting of one tension sample (Fig. 1c.).

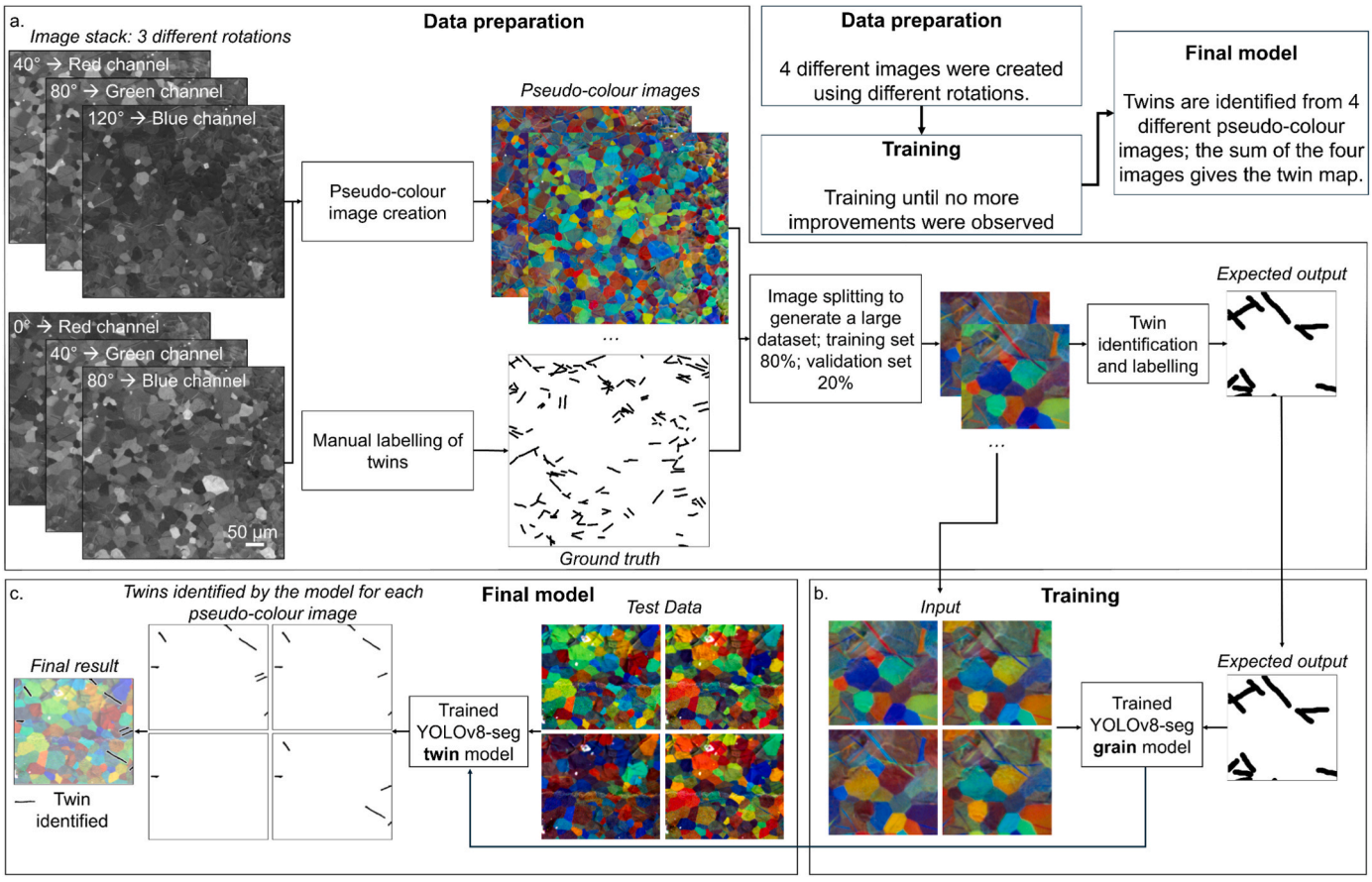


Fig. 1. Example of methodology to train a model YOLOv8-seg to recognize twinning regions: a. Using PLM images, four different 'pseudo-colour' images are created from three images at three different in-plane rotations (every 40°). These pseudo-colour images are then cropped into smaller areas. In total, 3144 images were assigned for training, and 792 images were assigned for validation. The ground truths (GT) are built from manual labelling of the twins. b. The training can start for several epochs. The model is trained from an already pre-trained YOLOv8-seg **grain** model. The trained model is saved and used for twinning regions identification in c. c. Finally, the model is tested. The trained YOLOv8-seg model identifies the different twinning regions in the pseudo-colour images.

2.3. Model implementation and training

The dataset and YOLOv8-seg model were developed using Python, employing the Ultralytics [23] and PyTorch [24] libraries. The YOLOv8 architecture is fundamentally composed of two principal components: the backbone, and the head, a configuration commonly adopted in modern object detection frameworks [25]. The accuracy and computational efficiency of YOLOv8 can be attributed to its dual parallel gradient flow branches, which promote a more effective propagation of gradient information throughout the network. The architecture alternates between convolutional layers and C2f modules, finishing in a Spatial Pyramid Pooling Fusion (SPPF) module. This final component enables the extraction of features at multiple spatial scales, thereby enhancing the model's capacity for generalization and its robustness in identifying twins across images of varying pseudo-colour images [26].

In the present study, a pretrained model was utilized, the YOLOv8-seg grain model from Ref. [21] was trained to recognize twins using the nano configuration of YOLO. The lightweight YOLOv8-nano configuration counts approximately 3.4 million trainable parameters. The YOLOv8-seg variant integrates an additional segmentation head inspired by YOLACT [27], which facilitates pixel-level mask generation.

For all samples, the corresponding polarized images were spatially registered to ensure that each manually labelled twinning region could be directly correlated with its representation in the optical images. To appraise the robustness of the proposed approach, five additional models were trained using different variation of the datasets, with their outcomes presented in Fig. 4. PLM images were co-registered with the manual labels, superimposed, and cropped by factors of four and five, thereby increasing the number of images available for training and validation. The resulting cropped regions were randomly assigned to training and validation subsets, and each image was resized to 640 × 640 pixels using zero-padding to maintain aspect ratio consistency.

To enhance dataset diversity, image augmentation was applied by rotating each image and its corresponding grain boundary (GB) map by 90°, 180°, and 270°. The final dataset comprised 3144 training images and 792 validation images, corresponding to an 80/20 split. Following cropping and resizing, the image resolutions ranged from 0.17 µm/pixel to 0.35 µm/pixel. The augmentations were designed under the assumption that rotational transformations would improve the model's capacity to recognize grains of varying shapes and orientations.

Model training was conducted over 750 epochs, where each epoch represented one complete pass through the training and validation data. The model was initialized with pretrained parameters (from Ref. [21]) and trained using a learning rate of 10^{-2}. The results presented in this study correspond to the state of the model after 750 training epochs. Although the loss function began converging rapidly within the initial epochs, indicating the onset of overfitting, the model continued to exhibit performance improvements. The results obtained after 750 epochs were notably superior to those observed after 50 epochs, the point at which overfitting behaviour began to appear. Similar training dynamics were observed across all additional models, as detailed in the supplementary materials.

2.4. Methodology to classify twin/parent relationships from segmented twin regions

Once the YOLOV8-seg models segmented grains and twin regions, post-processing methods were implemented to classify the specific twinning system. Firstly, conditions on the allowable shape of individual twins were implemented. It is possible to have irregular apparent twin grain morphologies, if the twinning plane normal (TPN) is oriented with the measurement surface [28]. However, in most cases, the TPN will not be oriented with the measurement surface, and the resulting section view will have a lenticular morphology, characterised by a high aspect ratio. In this study, it is assumed that it is not possible for a twin to be curved and cannot have branches. When a twinning region segmented by the model is curved and has one branch, then it will be 'split' into two regions, where each region shape approximates a straight line. Secondly, where the model segments twinning regions with multiple branches, it is considered that there are multiple individual twins. Here, the region is iteratively split, so the regions have one branch and are not curved (Fig. 2b.). To associate a parent grain to a segmented twin (after required splitting), the twins were skeletonized to straight lines. Then the centroids of any adjacent (neighbouring) segmented grain regions were projected perpendicularly onto the straight lines. If the projected point falls within the twinning regions length, then the neighbour is considered as a potential parent.

To identify a twin/parent system, a twinning region must either have one parent grain region, or two potential segmented parent grain regions. The latter occurs where the twin grain is connected to two grain boundaries, sectioning the parent grain in the measurement plane.

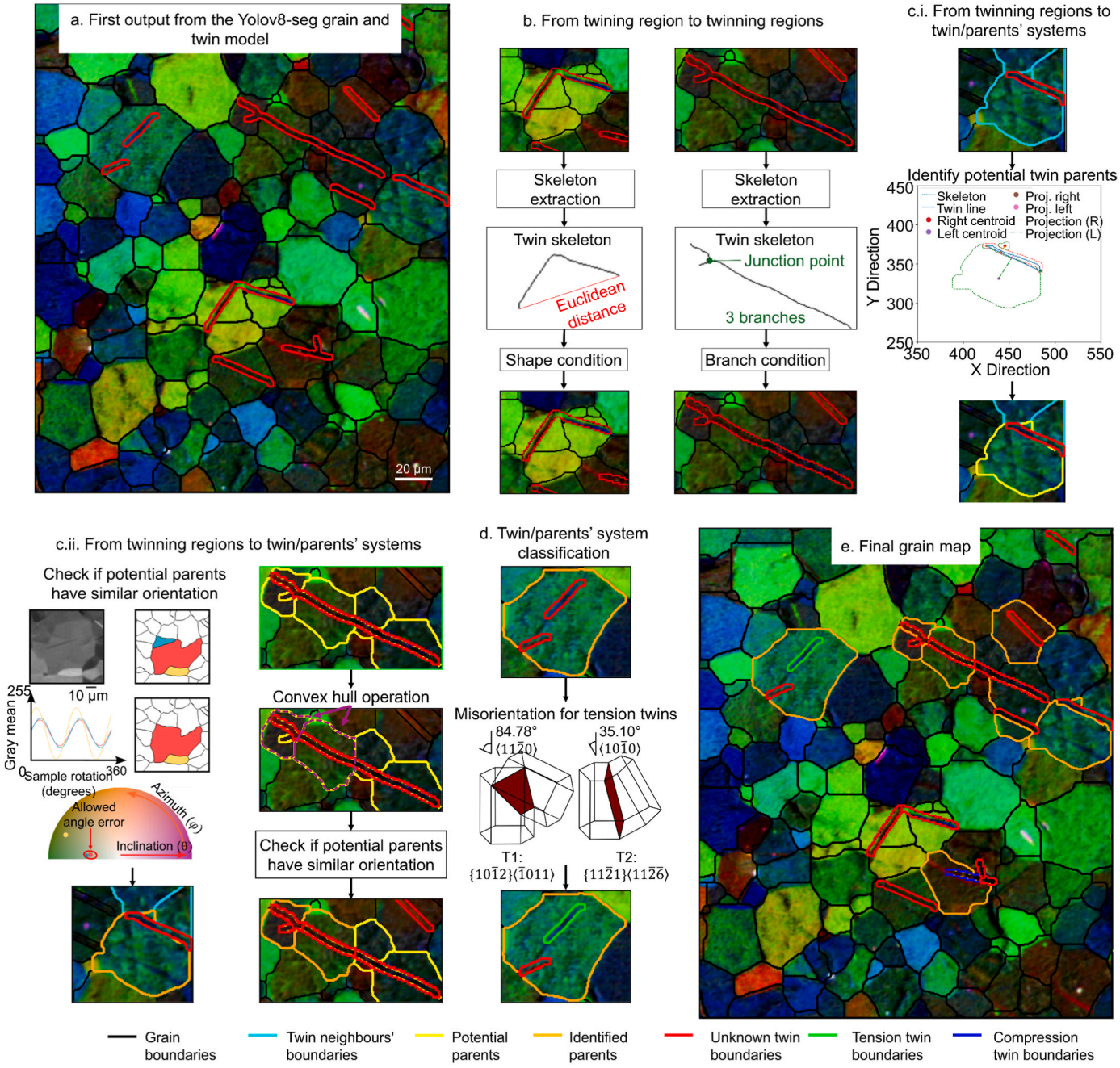


Fig. 2. Methodology to separate twinning regions to identify twins and their misorientation to their parents. Firstly, shape and branch conditions are used to cut twinning regions. Secondly, potential parents are identified from the neighbours of the twins. Then, twinning regions with too many parents are split if they have parents.

Where two grain regions are identified, they must have similar azimuth and inclination angles (Fig. 2c.). There are some cases where a twinning region has multiple potential parents on its side. In this case, some systems can be isolated based on their relative distances and using a convex hull operation [29]. Nevertheless, as shown in Fig. 7a., the post-processing phase fails to classify all twin/parents systems and more robust ML models would perform better.

2.5. Twin classification workflow with PLM

Once the twin/parents systems have been identified, the twin classification process can start (Fig. 3a.). With EBSD, in general, twins and parents are found measuring the misorientation between every individual grain and its neighbours. To determine the misorientation between the grains, orientation matrices are defined from the Euler angles $[\varphi, \theta, \phi]$ respectively: the azimuth angle, the inclination angle, and the rotation angle of the grain:

$$g_{grain} = \begin{pmatrix} c_\varphi c_\phi - s_\varphi s_\phi c_\theta & s_\varphi c_\phi + c_\varphi s_\phi c_\theta & s_\phi s_\theta \\ -c_\varphi s_\phi - s_\varphi c_\phi c_\theta & -s_\varphi c_\phi + c_\varphi s_\phi c_\theta & c_\phi s_\theta \\ s_\varphi s_\theta & -c_\varphi s_\theta & c_\theta \end{pmatrix} \quad (1)$$

The orientation matrix of the neighbouring grains can be defined the same way with the Euler angles. The misorientation matrix of a twin/parent relationship can then be measured using the equation below:

$$M = g_{twin} g_{parent}^{-1} \quad (2)$$

Equation (2) measures the misorientation with g_{twin} the orientation of the twin with the angles $[\varphi_t, \theta_t, \phi_t]$ and g_{parent} the orientation of the parent with the angles $[\varphi_p, \theta_p, \phi_p]$. From the misorientation matrix, the misorientation axis $\vec{r}$ (r_1, r_2, r_3) and angle Θ can be determined [30]:

$$\cos(\Theta) = \frac{M_{11} + M_{22} + M_{33} - 1}{2} \quad (3)$$

$$r_1 = \frac{M_{23} - M_{32}}{2 \sin(\Theta)}, r_2 = \frac{M_{31} - M_{13}}{2 \sin(\Theta)}, r_3 = \frac{M_{12} - M_{21}}{2 \sin(\Theta)} \quad (4)$$

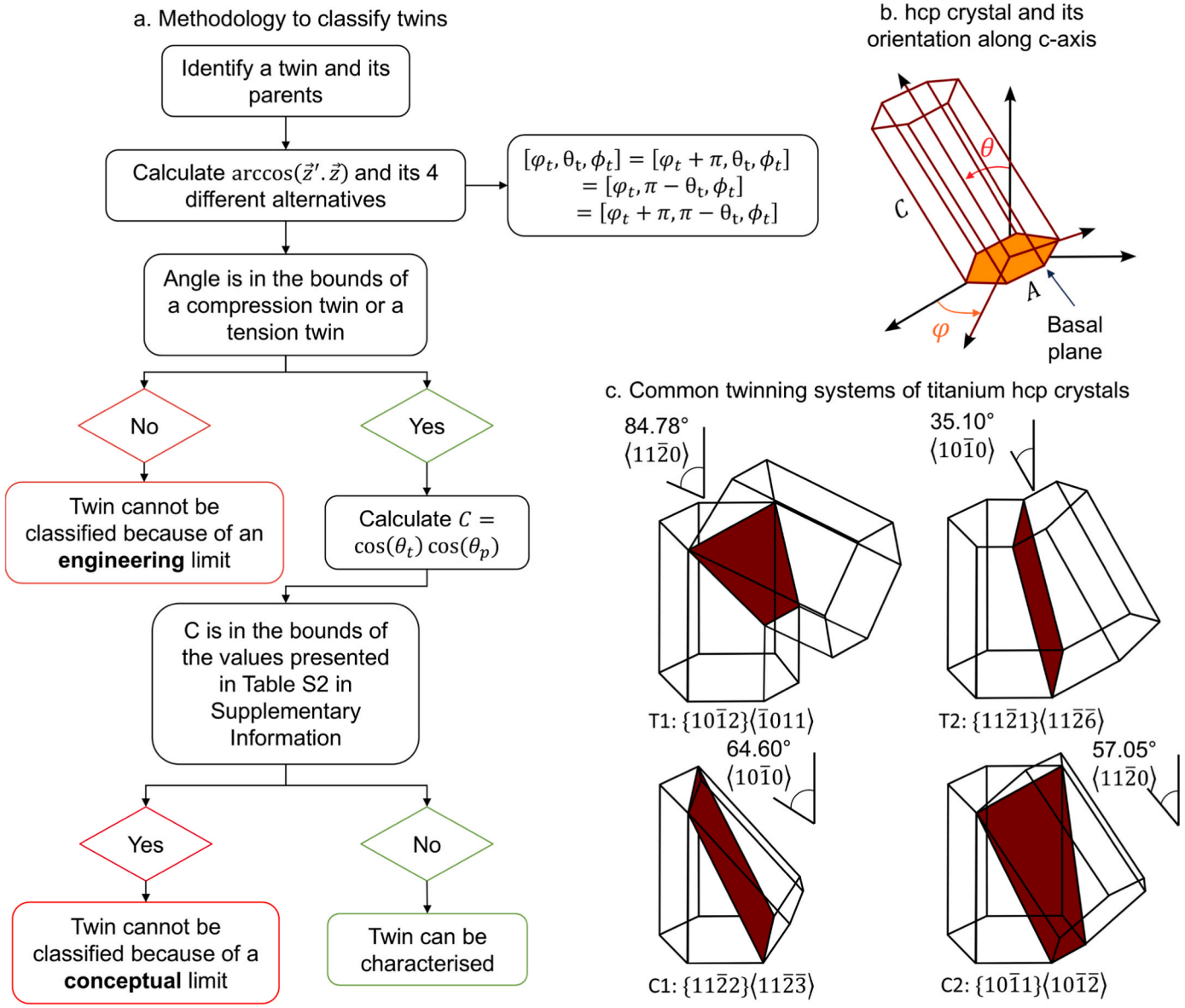


Fig. 3. Twin classification method: a. Methodology to classify twins; b. Hcp crystal with azimuth and inclination angles and C axis. c. Common twinning systems for titanium hcp crystals.

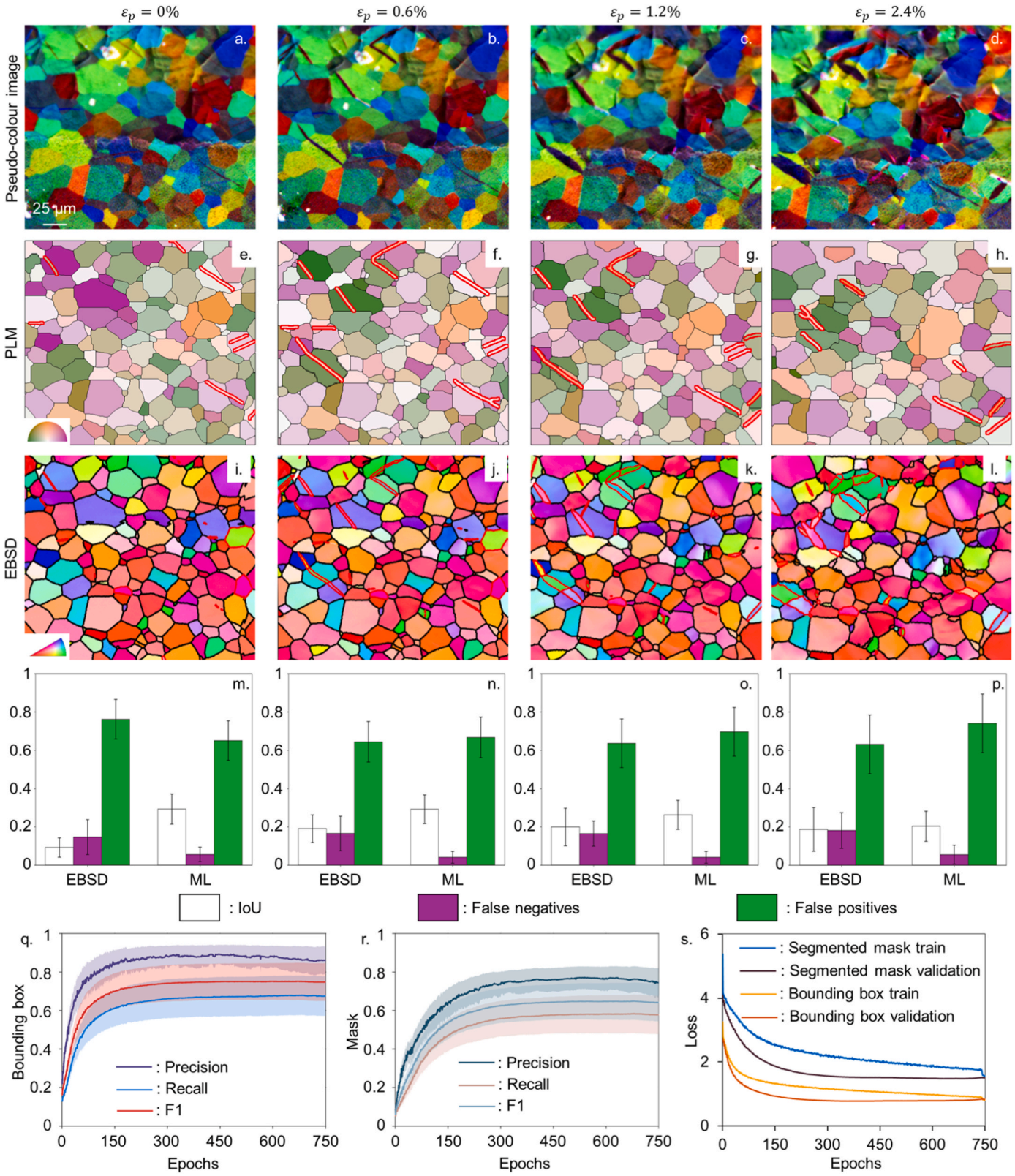


Fig. 4. Model performance for different levels of tensile plastic strain. For a tensile plastic strain from 0 to 2.4%, the twin identification model identify more twins than EBSD mapping for a step size and pixel size of 0.6 µm: a., b., c. And d. Represent the pseudo-colour images for a tensile plastic strain of 0%, 0.6%, 1.2% and 2.4%, e., f., g. And h. The PLM maps, i., j., k. And l. The EBSD maps, m., n., o. And p. The IoU, the false negatives and the false positives scores. Finally, q., r., and s. Represent the results for the different training sets: q. Bounding box scores for 750 epochs; r. Segmented mask scores for 750 epochs and s. Typical loss curves obtained during training.

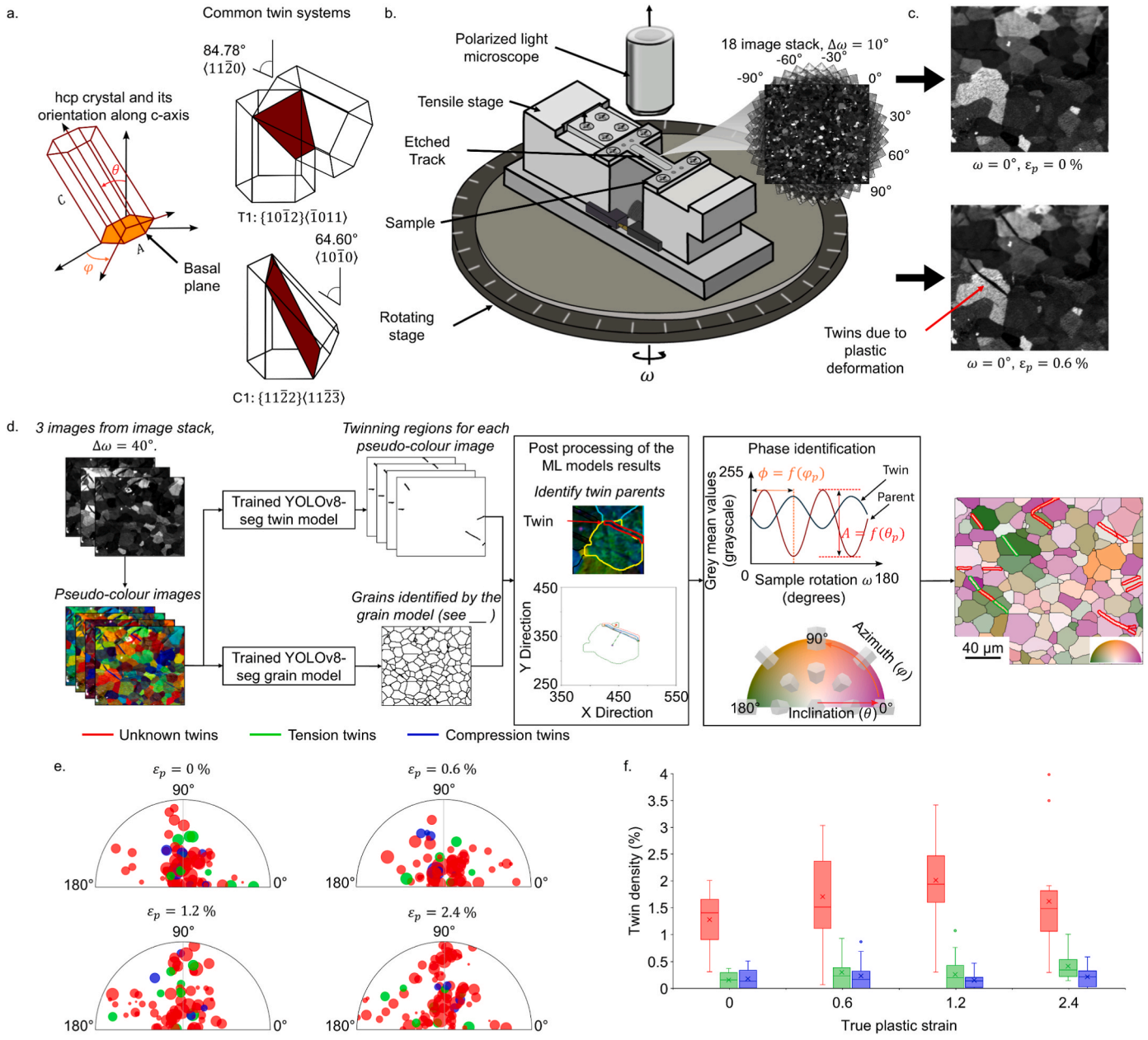


Fig. 5. A non-vacuum approach for twin segmentation and classification. a. α-Ti structure and two common twinning relationship (tension and compression). b. Tensile stage was mounted under a polarized light microscope used to capture images (in-plane rotations, ω, every 10°) for hcp materials (e.g. Ti, Mg, Zr). c. Twins appear in the sample upon plastic deformation. d. Workflow showing the two models used to segment grains and twins, respectively. After post-processing, PLM maps with identified twins are generated. e. Representative results from a tensioned sample with plastic strain, ε_p, ranging from 0, 0.6, 1.2 and 2.4%, showing pole figures of twins along the optical axis. f. Bar plots of the distribution of identified twins for the sample.

If the misorientation axis $\vec{r}$ and the angle Θ correspond to one of the systems in Table 1, then the twin can be classified. Nevertheless, it is common for chance grain boundary segments to meet a twin criterion, despite the grain map suggesting otherwise. Using the PLM approach, the workflow is different. Firstly, twins are directly segmented (based on shape) by the YOLOv8-seg twin model. Then, parents must be determined using geometrical rules (Fig. 2) as the obtained misorientation between a twin and its neighbours has too many solutions. Effectively, when the values of azimuth and inclination are determined using PLM, there are four different possible orientation solutions (Equation (5)), and the final misorientation between the parent and the twin could take 16 distinct values, challenging twin classification. Additionally, the rotation angle is unknown.

$$[\varphi_t, \theta_t, \phi_t] = [\varphi_t + \pi, \theta_t, \phi_t] = [\varphi_t, \pi - \theta_t, \phi_t] = [\varphi_t + \pi, \pi - \theta_t, \phi_t] \quad (5)$$

Morales-Rivas et al. [31] reported that, for α-Ti, all rotation axes for each twinning system can be projected along the basal plane (*see* Fig. 5a). Hence, the misorientation between the parent and a twin always consists of a pure rotation around a crystal axis normal to the C-axis of the crystal. If it is possible to determine the C-axis misalignment angle α between the twin and its parent, then it is possible to classify the twin relationship, as each misorientation angle is unique for each kind of twin (Table 1). This can be demonstrated by rewriting (2) into its other forms [30]:

$$g_{twin} = M g_{parent} \quad (6)$$

$$g_{twin} = g_{parent} + (1 - cos\,\alpha)\ \vec{n} \times \left(\vec{n} \times g_{parent}\right) + \sin\alpha\ \vec{n} \times g_{parent} \quad (7)$$

Then, the C-axis of both crystals, parent and twin, can be defined as unit vectors, $\vec{z}$ and $\vec{z}'$ and replaced in (7):

$$\vec{z}' = \vec{z} + (1 - cos\,\alpha)\ \vec{n} \times (\vec{n} \times \vec{z}) + \sin\alpha\ \vec{n} \times \vec{z} \quad (8)$$

$$\vec{z}' = \vec{z} + (1 - cos\,\alpha)\ (\vec{n}\ .(\vec{n}.\ \vec{z}) - \vec{z}.(\vec{n}.\vec{n})) + \sin\alpha\ \vec{n} \times \vec{z} \quad (9)$$

Here, rotation is only along the basal plane so $\vec{z}.\vec{n} = 0$ and $\vec{n}.\vec{n} = 1$:

$$\vec{z}' = \vec{z} - (1 - cos\,\alpha)\ \vec{z} + \sin\alpha\ \vec{n} \times \vec{z} \quad (10)$$

$$\vec{z}' = cos\,\alpha\ \vec{z} + \sin\alpha\ \vec{n} \times \vec{z} \quad (11)$$

$$\vec{z}'.\vec{z} = cos\,\alpha\ \vec{z}.\vec{z} + \sin\alpha\ (\vec{n} \times \vec{z}).\vec{z} \quad (12)$$

$\vec{n} \times \vec{z}$ is a vector perpendicular to $\vec{z}$ and so $(\vec{n} \times \vec{z}).\vec{z} = 0$ and $\vec{z}.\vec{z} = 1$ because $\vec{z}'$ and $\vec{z}$ are unit vectors, so finally:

$$\vec{z}'.\vec{z} = cos\,\alpha \quad (13)$$

Therefore, it is possible to classify a twin by measuring the angular difference of the optical angles between a twin and its parent. This will reduce the number of possible misorientation permutations to four, considering the solutions for the parent and the twin (see Supplementary Information, Table S1). Consequently, if one of the measured angles is close to one of the tension angles defined in Table 1 and the other measured angles are not close to compression angles, then the twin can be defined as a tension twin of a specified class, and vice versa.

3. Results & discussion

3.1. Performance of the YOLOv8-seg twin model

To be able to classify twins, it is necessary for both models: the YOLOV8-seg grain model and the YOLOV8-seg twin model, to accurately segment grains and twins. The performance of the YOLOV8-seg grain model has been discussed in previous work [21]. For the YOLOv8-seg twin model, training datasets were generated by manual labelling of pseudo-colour PLM images, as attempts to use EBSD-derived masks as automated ground truth produced insufficient label quality due to step size and pixel size limitations and misorientation criterion errors. It should therefore be noted that the performance metrics reported below reflect the model's consistency with expert PLM-based annotations, rather than an absolute crystallographic ground truth. The final trained YOLOv8-seg twin model was applied to the test sample across four levels of strain and compared to the corresponding EBSD dataset.

As it can be seen on the pseudo-colour image on Fig. 4a., several twins that can be observed (based on their morphologies with respect to the typical equiaxed microstructure) have not been segmented in the EBSD map on Fig. 4i. Additionally, several grains were wrongly labelled as twins when using only the misorientation condition with the EBSD dataset, an issue reported by Savage et al. [13]. However, more twins were identified using the YOLOV8-seg twin model for a plastic strain of $\varepsilon = 0\%$ (Fig. 4m.) without mistaking surface defects for twins. Effectively, the ratio of false positives (i.e. segmenting a twin region that has not been manually labelled) for the YOLOV8-seg is around 0.05 for the three levels of strains, indicating it rarely identifies surface defects at twins. On the other hand, the ratio of false positives for the EBSD dataset is three times higher (around 0.15) when only the misorientation criterion is used to identify twins. Even if the exact ratio of false positives when only using the misorientation criterion has not been measured, Wright et al. [32] reported some errors when reconstructing a grain size map without the twins, and in the final grain size distribution when only using the misorientation criterion. The ratio of false negatives (i.e. not segmenting a twin that has been manually labelled) is high for both EBSD maps and the YOLOV8-seg twin model. For the YOLOV8-seg twin model, the ratio of false negatives is increasing from 0.65 to 0.75 as the strain increases. This is likely due to the training dataset being composed of samples with a true plastic strain up to 1.0%. Consequently, the model struggles to capture twins as they enlarge at higher strain levels, as shown in Fig. 4o And p. On the contrary, the ratio of false negatives achieved using EBSD data decreases from 0.76 to 0.63, as it is easier to identify twins at a higher strain levels as twin size increases [33]. Finally, the intersection over union ratio (IoU), which indicates the capability of both methods to correctly identify the different twins, is higher for the YOLOV8-seg twin model than for the EBSD dataset. This ratio is three times higher when the sample is not strained (0.3 against 0.1 for the EBSD dataset). This advantage diminishes with increasing deformation, converging to comparable values of 0.20 and 0.19 respectively at a true strain of 2.4%. These results suggest that, relative to expert PLM annotations, the YOLOv8-seg twin model provides more consistent twin segmentation than a simple EBSD misorientation criterion at low strain levels, while EBSD-based methods may benefit from additional post-processing criteria to reduce false positives.

The model's performance relative to manual PLM annotations is strongest up to a plastic strain of 1.0%, beyond which twin growth challenges the model's generalisation. Additional training data at higher strain levels could improve robustness, though deformation beyond 1% in α-Ti alloys is generally of limited engineering interest due to associated surface buckling. In future work, EBSD datasets, generated with fine step sizes and pixel sizes and multi-criterion twin identification protocols, could serve as an independent, automated source of ground truth for training more advanced PLM-based models for grain segmentation, twin identification, and parent-twin recognition, thereby removing the potential circularity bias in the present manual labelling approach.

3.2. Non-vacuum approach to twin classification

Hence, thanks to the different trained models: YOLOv8-seg twin model and YOLOv8-seg grain model, twins and grains can be respectively identified on the sample and then classified (Fig. 5b.). The test sample was mounted on a mechanical testing stage to generate different levels of plastic strain ($\varepsilon_{plastic} = 0\%$, 0.6%, 1.2% and 2.4%) in tension (Fig. 5b.). Plastic deformation causes twinning as high plastic deformation favours twin nucleation [6] (Fig. 5c.). From the grain map and the twinning regions, PLM maps can be generated after post-processing of the results from the ML models (Fig. 5d.). As previously explained, with PLM, only 2 out of the 3 Euler angles can be identified: the azimuth and inclination angles, but not the rotation angle. Moreover, for the azimuth and inclination angles, only the orientation along the optical axis can be determined. The azimuth angle obtained will be between 0° and 180° (instead of 0° to 360°), and the inclination angle from 0° to 90° (instead of 0° to 180°). Compared to EBSD, PLM provides only one-third of the crystallographic information, specifically, two out of the three Euler angles, each limited to half of the angular range accessible by EBSD, making twin classification more challenging (Fig. 5d.). Hence, due to i) the inaccuracy of the trained models, ii) limitations of PLM and iii) the difficulty in identifying parent grains at high level of deformation, some twins are categorised as unknown when it was not possible to conclude on their types. It can be observed that the twin area density was increasing, on average, from 1.62% at $\varepsilon_{plastic} = 0\%$ to 2.42% at $\varepsilon_{plastic} = 1.2\%$ before decreasing to 2.25% at $\varepsilon_{plastic} = 2.4\%$. The density of tension twins increases at the plastic strain augments from 0.16 to 0.41% at $\varepsilon_{plastic} = 2.4\%$ which is to be expected as the plastic strain increases. On the other hand, the density of compression twins varies between 0.15% and 0.22%. The density of segmented twins that are not classified and remain unknown is high, representing 80% of the identified twins. In general, more variation appears at higher strain, like

$\varepsilon_{plastic}$ = 2.4%, because the grains are more deformed and the twins larger at this plastic strain level. This is likely a limitation of the training dataset that was composed of samples with a maximum level of plastic strain of 1.0%.

Fig. 6 demonstrates how to classify twins into a particular class. Fig. 6a shows a pseudo-colour image of a sample prior to loading, composed of three grayscale images acquired with a defined in-plane rotation ($\Delta\omega = 40°$). Fig. 6b shows the output of the panoptic segmentation models, where segmented grain regions are shown with black boundaries and segmented twin regions are shown with red boundaries. Deformation observed prior to tensile loading likely results from either the sheet rolling process or from mechanical polishing (these processes could introduce compressive strain into the material). Fig. 6c shows the resulting C-axis orientation each segmented region, calculated following the methodology described in previous work [20]. This process is repeated on the same sample after tensile loading to $\varepsilon_{plastic} = 0.6\%$ (Fig. 6d-f). A new twin is formed upon sample extension (labelled in Fig. 6d); it can be deduced that this twin will fall into one of the tensile twin systems. Fig. 6e shows the segmentation result, and Fig. 6f shows the corresponding C-axis orientation map.

C-axis orientations can be determined using equation (A.2) and (A.3) for the 'new' twin formed during loading ($\varphi_t = 93.2°$, $\theta_t = 45.2°$), and its parent grain ($\varphi_p = 176.9°$, $\theta_p = 77.5°$). Using equation (13), four unique solutions for the angular difference between the two C-axes can be measured using the following equation:

$$\alpha = \arccos\left(\vec{z}'.\vec{z}\right) \tag{14}$$

The four unique angular difference solutions are: 76.7°, 85.7°, 94.3°

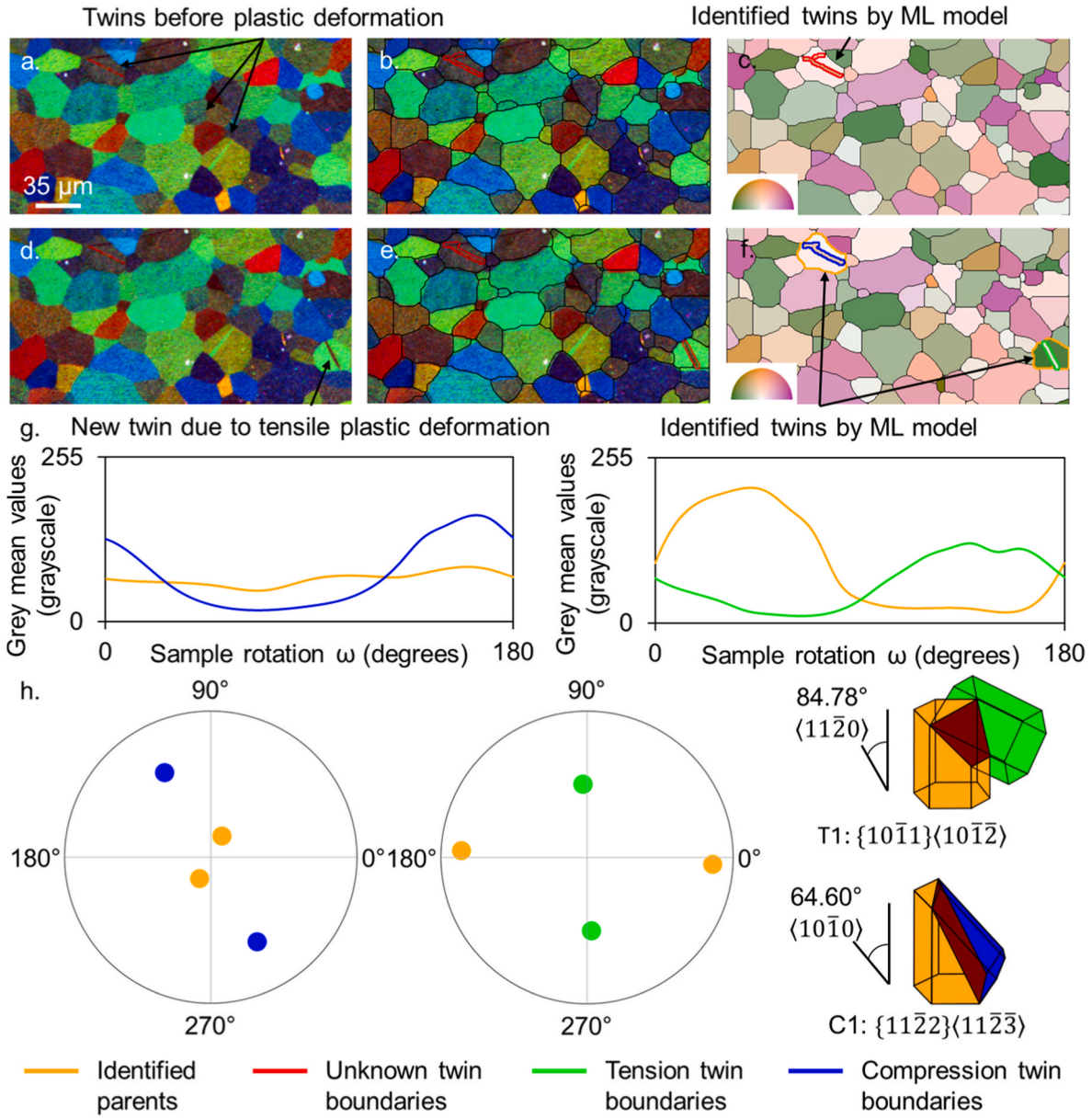


Fig. 6. Deformation twin classification in hcp materials using PLM, panoptic segmentation, and fundamental crystallographic principles. a. Pseudo-colour image of unloaded sample in the as-received condition ($\varepsilon_{plastic} = 0\%$). b. Pseudo-colour image with overlaid region boundaries for grains (black) and twins (red), segmented by the trained models. c. C-axis orientation map for the unloaded sample using segmentation in b. d. Pseudo-colour image of tensile loaded sample ($\varepsilon_{plastic} = 0.6\%$), and e. Overlaid with segmented region boundaries for grains and twins. f. C-axis orientation map for the unloaded sample using segmentation in e. g. Extracted traces from the labelled pre-existing twin and its parent grain, and h. The same for the extension twin formed upon loading. Pole figures for the two identified twins: a compression twin (likely due to sample preparation or rolling), and a tensile twin formed upon loading.

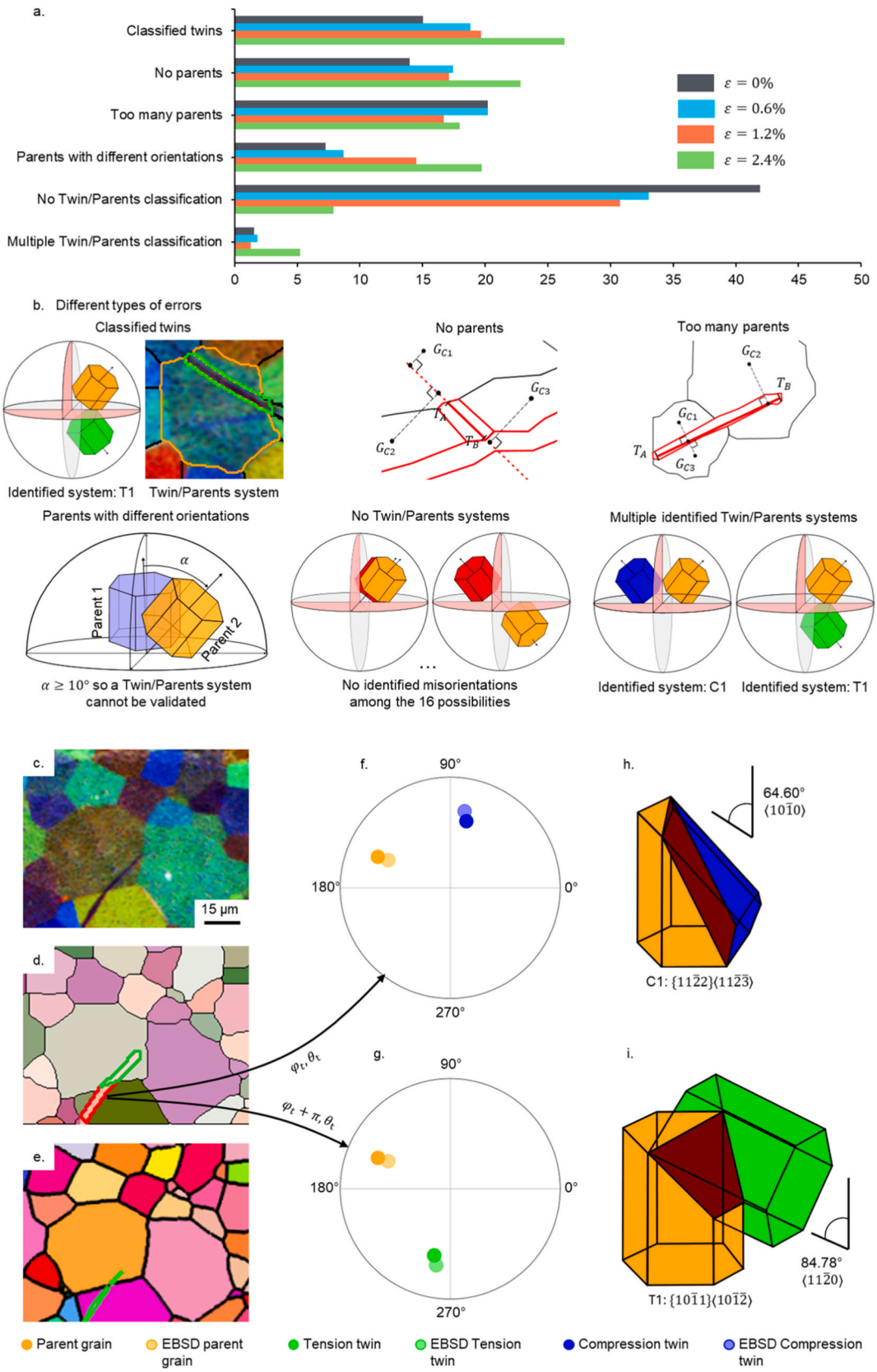


Fig. 7. Error sources using the PLM-ML approach. a. Bar plot of different errors that are occurring during the classification process; b. Illustration of the limitations of PLM for classifying twins; c. Pseudo-colour image; d. PLM map; e. EBSD Map; f. Pole figure for the first case, showing a compression twin identified by PLM; g. Pole figure for the second case, when π is added to the twin azimuth value; h. And i. Represent the different crystals.

and 103.3°. 85.7° is within the accepted error range (±5°) to 84.8°, the angle for a tension twin: $\{10\bar{1}2\}\langle\bar{1}011\rangle$. Similarly, for the pre-existing twin segmented at both levels of strain $\varepsilon_{plastic} = 0\%$ and $\varepsilon_{plastic} = 0.6\%$, the same classification process can be followed, to output four possible angular difference solutions between the parent and twin: 52.0°, 68.3°, 111.7° and 128.0°. 68.3° falls within the accepted range of 64.60° (±5°), therefore the twin is classified as a compression twin: $\{11\bar{2}2\}\langle11\bar{2}3\rangle$. Following this methodology of combining PLM with an accessible panoptic segmentation model and the application of fundamental crystallographic rules, twin systems can be segmented and classified, despite the reduced amount of accessible crystallographic information.

As shown in Fig. 5f., the proportion of unknown twins (i.e. segmented, but unclassified) is approximately 80% in the current implementation. As detailed in Fig. 7 and discussed below, this high proportion is predominantly attributable to engineering limitations: including segmentation errors, parent grain association failures, and azimuth angle measurement imprecision, rather than to fundamental theoretical constraints inherent to PLM, which account for approximately 5% of cases. Error sources that lead to an 'unknown' response are outlined in Fig. 7b.

1. No parent grains are associated to a twin.
2. Too many parents (two or more on one side) are associated to a twin.
3. The associated parent grain sections to a twin have inconsistent orientations (i.e. are not the same grain).
4. No solution for the twin/parent system for α-Ti is determined within the allowable misorientation error (±5°).
5. Multiple (conflicting) solutions for the twin/parent system for α-Ti (e.g. compression and tension) within the allowable misorientation error (±5°).

The first two errors primarily result from image processing issues linked to the ML models. In this study, two accessible YOLOv8-seg models were trained to segment parent grains and twins using a limited number of samples subjected to limited tensile plastic strain ($\varepsilon_{plastic} < 1.0\%$). Consequently, the segmentation accuracy for both grains and twins is restricted. Additionally, the geometrical approach used to identify parent grains can introduce further errors. Improvements could involve training ML models to directly recognize parent–twin relationships. These segmentation-related errors represent 35% to 40% (34% at $\varepsilon_{plastic} = 0\%$, 38% at $\varepsilon_{plastic} = 0.6\%$, 34% at $\varepsilon_{plastic} = 1.2\%$ and 41% at $\varepsilon_{plastic} = 2.4\%$) of all identified twins and could be mitigated using more advanced ML models and/or larger training datasets.

Another frequent issue is the occurrence of twins with two parents with dissimilar orientations (7% at $\varepsilon_{plastic} = 0\%$, 9% at $\varepsilon_{plastic} = 0.6\%$, 15% at $\varepsilon_{plastic} = 1.2\%$ and 20% at $\varepsilon_{plastic} = 2.4\%$). Segmented parent grain regions should share a consistent twin/parent relationship. This error arises from segmentation inaccuracies and from measurement uncertainties in azimuth and inclination angles using PLM, detailed by Ahmed et al. [20]. This error increases from 7% to 20% with rising plastic strain, largely due to the increasing segmentation difficulty in highly strained regions, where the surface starts to buckle. The most prevalent error, however, is no solution to a known twin/parent system for α-Ti, which represents 41% of the identified twins at $\varepsilon_{plastic} = 0\%$, 30%, at $\varepsilon_{plastic} = 1.2\%$ but decreases to 8% at $\varepsilon_{plastic} = 2.4\%$. This error originates from a combination of segmentation inaccuracies, PLM measurement imprecision, and post-processing limitations. Here, the angle between the twin and parent grain does not match any of the expected values listed in Table 1. A final, less frequent but particularly ambiguous case occurs when a twin is simultaneously identified as both a tension and a compression twin (conflicting solutions for the allowable misorientations). This situation, observed in less than 5% of cases, reflects a fundamental limitation of the method, as each twin/parent system has four potential misorientation solutions. Two of these may fall into distinct twin/parent systems (one compression, one tension).

Overall, PLM demonstrates potential for twin/parent system classification, subject to both engineering and theoretical constraints. The engineering limitations, which currently represent for most unclassified twins, are in principle addressable through more robust ML models, larger and more diverse training datasets, improved azimuth and inclination angle calibration, and automated parent-twin recognition. Theoretical limitations, arising from cases where PLM cannot distinguish between compression and tension twin systems due to the reduced crystallographic information available, represent a smaller constraint that must be accounted for in any future implementation.

In Fig. 7b., a twin is classified as 'unknown' by the method because it is alternately identified as a compression and a tension twin depending on the tested configuration. Using the PLM approach, the parent has an azimuth of 158.0° (156.0° according to EBSD) and an inclination of 64.6° (55.7° according to EBSD), while the twin has an azimuth of 76.4° (79.4° according to EBSD) and an inclination of 55.8° (63.5° according to EBSD). Using the dot product, the angle α obtained between the two C-axes is 69.5° (65.0° according to EBSD), which indicates a possible compression twin: $\{11\bar{2}2\}\langle11\bar{2}3\rangle$, within allowable error range (±5°).

To determine whether this case could lead to conflicting solutions, the value of C can be evaluated by $\cos(\theta_t)\cos(\theta_p)$, which in this case is equal to 0.241. Supplementary Information Table S2 shows that 0.241 is between the lower and upper bounds of $C = (a + b)/2$, when the other possibility is a tension twin with an angle of 84.8°. Fig. 7f Shows that another combination ($\phi_{1p} + \pi$, Φ) allows an additional solution of 82.4°, giving two possible solutions that conflict. EBSD however, identified this twin as a tension twin (Fig. 7e.). Thus, while classifying twins using PLM datasets is feasible, the reduced amount of crystallographic information can impose fundamental limitations. To minimize the likelihood of identifying a twin as both a compression and a tension twin, the angular tolerance (±5°) would need to be reduced. However, this tolerance was originally chosen to offset the influence of inclination measurement errors in PLM [19]. Some studies have shown that the precision of PLM for measuring inclination angles can be improved, notably by calibrating the inclination values with EBSD values, and this will further enhance the technique's ability to directly classify twins in hcp materials [17,18].

In conclusion, this study demonstrates that PLM, when combined with panoptic segmentation and crystallographic rules-based post-processing, can be an effective approach to allow the segmentation and classification of twins and thus enable the analysis of the plastic deformation history of materials. Even though PLM captures only about one-third of the crystal orientation data compared with EBSD, it can classify some twins in α-Ti samples. The proportion of unclassified twins in the current implementation remains significant, primarily due to engineering limitations such as segmentation accuracy and azimuth angle measurement precision, rather than fundamental theoretical constraints of the technique, which account for approximately 5% of cases.

3.3. Future materials digitalization using PLM

Although diffraction-based techniques, i.e. EBSD, will always be essential for fundamental microstructural analysis, they are not practical for in-line manufacturing part characterisation or digital manufacturing workflows. Several technologies have been explored for automated materials characterisation and fast microstructure control (directional reflectance microscopy [34], spatially-resolved acoustic spectroscopy [35], topography orientation imaging [36]). The present work demonstrates new capabilities that enable PLM to function as a direct, production-ready materials characterisation method (Fig. 8). PLM has been applied to petrographic characterisation as well as for optically anisotropic alloy systems [37]. With the progress of digital technologies including image processing with machine learning, large dataset management and digital image correlation, some of limitations of PLM can be overcome to allow the evaluation of metals in a production setting

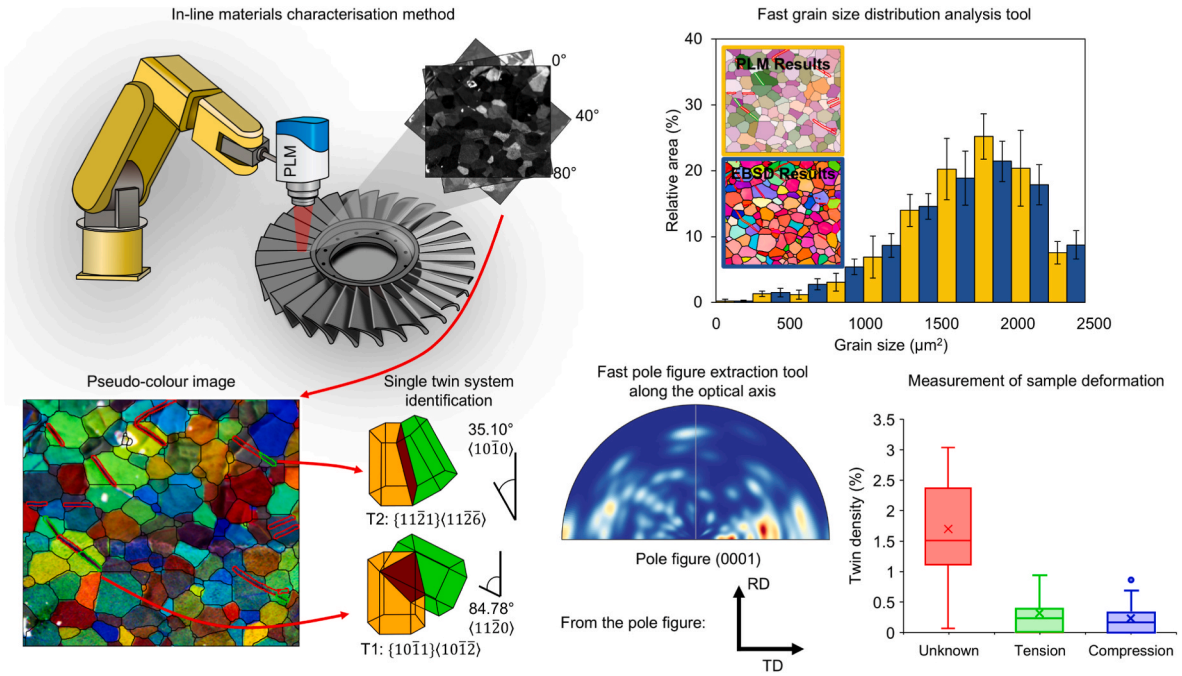


Fig. 8. With the development of new digital methods, PLM can be used as a direct materials characterisation method for grain size distribution analysis, fast material texture analysis and now for single twin system identification but more important part deformation.

[38]. PLM has been demonstrated to determine grain distributions [21], construct three-dimensional microstructure datasets [20], and now to segment and classify deformation defects. If this research uses manual dataset as the ground truth to identify twins as a first approach, automatic dataset could be built in the future using EBSD methods as diffraction-based techniques are more complete and will always be more accurate to identify twins and give indications on crystal orientations.

4. Conclusion

This work has demonstrated the potential of PLM for deformation twin segmentation and partial classification in hcp Ti alloys. Although PLM recovers only a third of the crystallographic information accessible via diffraction-based methods, specifically the c-axis azimuth and inclination angles, but not the rotation angle, this information is in principle sufficient to classify twins in hcp materials provided that theoretical and engineering constraints are managed appropriately. The same approach could be extended to other hcp systems including Mg, Co, and Zr alloys, whose twinning systems are similarly defined by rotations about the basal plane.

Twin classification using PLM is subject to two categories of limitation. Theoretical limitations represent a constraint inherent to the technique. In the present study, these account for approximately 5% of unclassified twins. Engineering limitations, including segmentation errors, uncalibrated azimuth angle measurements, and imprecise parent-twin association, represent the dominant source of unclassified twins in the current implementation but could decrease through future developments. With improved ML models trained on larger and more diverse datasets, calibrated PLM acquisition systems, and automated parent-twin recognition algorithms, the proportion of classifiable twins is expected to increase.

PLM was additionally shown to provide more consistent twin segmentation than EBSD using a simple misorientation criterion at low strain levels, within the scope of the tested samples. Taken together, these results position PLM combined with machine learning as a practical, low-cost, and vacuum-free foundation for deformation characterisation in manufacturing environments, complementary to rather than a replacement for EBSD in applications requiring full crystallographic orientation data.

Code availability

The code required to reproduce these findings is available to download from https://github.com/girerdth/PLM_ML_Twins_Classification. Additionally, the last trained models (for twin and grain identification) are available to download in the repository.

CRediT authorship contribution statement

Thomas Girerd: Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing – original draft. **Giovanna Martinez-Arellano:** Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Adam Thomas Clare:** Funding acquisition, Methodology, Supervision, Validation, Writing – review & editing. **Alistair Speidel:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.mtadv.2026.100841.

Data availability

Data will be made available on request.

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